1. I2S0/TDM PROGRAM–AUDIO LVDS LINK

J901 mates with AUDIO−8X8 J101. Route five 100 ohm differential pairs with the assigned interleaved returns.

SN65LVDT2D includes the ADC−data receiver termination; do not add a duplicate far−end resistor.

2. DEDICATED LOW–NOISE HEADSET RAILS

LP5907 rails are dedicated to the codec/headphone path. HEADSET_POWER_EN comes from the GPIO expander.

3. ES8316 I2S1 HEADSET CODEC

Codec pinout and baseline decoupling follow Radxa CM5 IO V2.2; validate Everest production datasheet values before release.

4. TPA6132A2 HEADPHONE AMPLIFIER

G0=0 and G1=1 strap the amplifier for 0 dB gain. EN remains software−controlled for pop−free mute.

5. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD

CTIA mapping is locked. The PJ−327A supplier footprint still requires mechanical verification before PCB release.

6. HEADSET I2C AND CONTROL POLICY

ES8316 is controlled on HS_I2C_SCL/HS_I2C_SDA. No GPIO−expander interrupt is required; status is polled.

HEADSET_POWER_EN and HEADSET_AMP_EN default low through 100k pulldowns so the jack remains muted during boot.

Sequence: enable codec rails, configure ES8316, enable TPA6132A2, then ramp software volume.

PJ−327A is wired CTIA: tip left, ring 1 right, ring 2 ground, sleeve microphone
TPA6132A2 provides the requested dedicated stereo headphone amplification
I2S1 drives an ES8316 headset codec based on the Radxa CM5 IO V2.2 reference
I2S0/TDM program audio is buffered over LVDS to the separate AUDIO−8X8 board
ProComm
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Title: Radxa CM5 ProComm Carrier − TDM and Headset Audio
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